

Finding error

```
1 #include <stdio.h>
2
3 int add(int a, int b);
4 float average(int a, int b, int c);
5
6 int main()
7 {
8     int age = "Twenty";
9     // Error: Assigning a string literal to an int variable.
10    // Solution: int age = 20;
11
12    float salary = 50000.50;
13
14    int marks = 85;
15
16    char grade = "A";
17    // Error: Assigning a string literal to a char variable.
18    // Solution: char grade = 'A';
19
20    printf("%d\n", salary);
21    // Warning: Incorrect format specifier for float variable.
22    // Solution: printf("%f\n", salary);
23
24    printf("%f\n", marks);
25    // Warning: Incorrect format specifier for int variable.
26    // Solution: printf("%d\n", marks);
```

```
27
28  int total;
29
30  total = marks + score;
31  // Error: Variable 'score' is undeclared.
32  // Solution: Declare score before use or replace with a valid variable.
33
34  int x = 10;
35  int x = 20;
36  // Error: Redclaration of variable x in the same scope.
37  // Solution: Use a different variable name or remove the duplicate declaration.
38
39  100 = x;
40  // Error: Cannot assign a value to a constant.
41  // Solution: x = 100;
42
43  int num = 25;
44
45  printf("%s\n", num);
46  // Warning: Incorrect format specifier for int variable.
47  // Solution: printf("%d\n", num);
48
49  float pi = 3.14;
50
51  printf("%d\n", pi);
52  // Warning: Incorrect format specifier for float variable.
53  // Solution: printf("%f\n", pi);
54
```

```
55  int arr[5];
56
57  arr[2.5] = 100;
58  // Error: Array index must be an integer expression.
59  // Solution: arr[2] = 100;
60
61  char ch = 65.5;
62  // Warning: Implicit conversion from float to char may lose data.
63  // Solution: char ch = 65;
64
65  int result;
66
67  result = add(10);
68  // Error: Function add() requires two arguments but only one is provided.
69  // Solution: result = add(10, 20);
70
71  printf("%d\n", result);
72
73  float avg;
74
75  avg = average(10,20);
76  // Error: Function average() requires three arguments but only two are provided.
77  // Solution: avg = average(10, 20, 30);
78
79  printf("%f\n", avg);
80
81  int temperature = "35";
82  // Error: Assigning a string literal to an int variable.
```

```
83 // Solution: int temperature = 35;
84
85 int amount = 1000;
86
87 float discount = "10.5";
88 // Error: Assigning a string literal to a float variable.
89 // Solution: float discount = 10.5;
90
91 amount = amount - discount;
92 // Warning: Implicit conversion from float to int may lose precision.
93 // Solution: Use float amount or cast explicitly.
94
95 char name[20];
96
97 name = "Robil";
98 // Error: Cannot assign a string literal directly to a character array.
99 // Solution: strcpy(name, "Robil");
100
101 printf("%s\n", name);
102
103 int count;
104
105 count = totalMarks + 10;
106 // Error: Variable 'totalMarks' is undeclared.
107 // Solution: Declare totalMarks before use.
108
109 int number = 50;
110
```

```
111  printf("%c\n", number);
112  // Warning: Printing an integer as a character may be unintended.
113  // Solution: printf("%d\n", number);
114
115  float rate = 7.5;
116
117  printf("%d\n", rate);
118  // Warning: Incorrect format specifier for float variable.
119  // Solution: printf("%f\n", rate);
120
121  int a = 10;
122  int b = 20;
123
124  int sum;
125
126  sum = add(a,b,c);
127  // Error: Function add() accepts only two arguments and variable c is undeclared.
128  // Solution: sum = add(a, b);
129
130  printf("%d\n", sum);
131
132  float radius = 5.5;
133
134  int area;
135
136  area = 3.14 * radius * radius;
137  // Warning: Assigning floating-point result to int causes loss of precision.
138  // Solution: float area = 3.14 * radius * radius;
```

```
139
140 printf("%d\n", area);
141
142 int studentMarks = 90;
143
144 if(studentMarks = 100)
145 // Error: Assignment used instead of comparison.
146 // Solution: if(studentMarks == 100)
147 {
148     printf("Perfect Score\n");
149 }
150
151 int size = "Large";
152 // Error: Assigning a string literal to an int variable.
153 // Solution: Use a string variable: char size[] = "Large";
154
155 printf("%d\n", size);
156
157 float percentage = 89.75;
158
159 printf("%d\n", percentage);
160 // Warning: Incorrect format specifier for float variable.
161 // Solution: printf("%f\n", percentage);
162
163 int arr2[10];
164
165 arr2["five"] = 100;
166 // Error: Array index must be an integer expression.
```

```
167 // Solution: arr2[5] = 100;
168
169 int choice = 2;
170
171 switch(choice)
172 {
173     case 1:
174         printf("One\n");
175         break;
176
177     case "2":
178         // Error: Case label must be an integer constant, not a string literal.
179         // Solution: case 2:
180         printf("Two\n");
181         break;
182 }
183
184 int finalResult;
185
186 finalResult = multiply(10,20);
187 // Error: Function multiply() is undeclared and undefined.
188 // Solution: Define multiply() or replace with a valid function.
189
190 printf("%d\n", finalResult);
191
192 int length = 10;
193 int width = 5;
194
```

```
195 float areaRect;
196
197 areaRect = length * width;
198
199 printf("%d\n", areaRect);
200 // Warning: Incorrect format specifier for float variable.
201 // Solution: printf("%f\n", areaRect);
202
203 int marks1 = 70;
204 int marks2 = 80;
205 int marks3 = 90;
206
207 int avgMarks;
208
209 avgMarks = average(marks1, marks2, marks3);
210 // Warning: Assigning float return value to int causes loss of precision.
211 // Solution: float avgMarks = average(marks1, marks2, marks3);
212
213 printf("%d\n", avgMarks);
214
215 int data;
216
217 printf("%d\n", value);
218 // Error: Variable 'value' is undeclared.
219 // Solution: Declare value before use or replace with a valid variable.
220
221 int employeeId = "EMP101";
222 // Error: Assigning a string literal to an int variable.
```



```
223 // Solution: char employeeId[] = "EMP101";
224
225 printf("%d\n", employeeId);
226 // Warning: Incorrect format specifier if employeeId is intended to be a string.
227 // Solution: printf("%s\n", employeeId);
228
229 return 0;
230 }
231
232 int add(int a, int b)
233 {
234     return a + b;
235 }
236
237 float average(int a, int b, int c)
238 {
239     return (a + b + c) / 3.0;
240 }
```